

MARGARET GATHONI

Data & Software Engineer | Analytics Systems | Backend & ETL
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PROFESSIONAL SUMMARY

Data & Software Engineer with 6+ years of experience designing data pipelines, analytical systems, and backend services across healthcare, NGO, and enterprise environments. Proven ability to bridge analytics and engineering, building ETL pipelines, APIs, dashboards, and automated reporting systems that translate complex data into actionable insights. Strong in Python, SQL, FastAPI/Flask, cloud workflows, and BI tools, with leadership experience at national program level.

CORE SKILLS

Engineering & Systems

- Python, FastAPI, Flask
- REST APIs, Microservices
- PostgreSQL, MySQL, Oracle
- Docker, AWS, CI/CD (GitHub Actions)
- ETL Architecture & Automation

Data & Analytics

- Data Modeling & SQL Analytics
- Power BI, Tableau, Looker
- Statistical Analysis (Python, R)
- Data Quality Audits & Validation
- Record Linkage & Text Matching
- Feature Engineering

PROFESSIONAL EXPERIENCE

CIHEB Kenya — HMIS Specialist | 2025

- Designed analytical schemas for 15+ health facilities, improving consistency and data reliability.
- Built Python-based QA and validation automation tools.
- Trained 200+ staff on digital systems, analytics, and data use.

CIHEB Kenya — Data Lead / ETL Engineer | 2024 – 2025

- Led national-level clinical data systems using PostgreSQL and MySQL.
- Designed and deployed Python ETL pipelines for automated reporting and surveillance.
- Built Power BI dashboards supporting monitoring, publications, and decision-making.
- Implemented automated data-quality audits, reducing errors by 30%.

Jay & Oak Ltd — Python Developer (Data Systems & APIs) | 2023 – 2024

- Built backend APIs and analytics systems using FastAPI, PostgreSQL, Docker.
- Developed data pipelines, cleaning scripts, and matching algorithms.
- Improved record-linkage accuracy by 40% and reduced processing time by 50%.
- Implemented CI/CD pipelines and automated integrations.

Aga Khan University Hospital — Research Data & Automation Analyst | 2016 – 2022

- Analyzed **100,000+ clinical and laboratory records** for research studies.
- Automated statistical reporting using Python and R.
- Built reproducible analytics scripts and dashboards for research teams.

Kaggle — Data Scientist Mentee (BIPOC Program) | 2022 – 2023

- Completed hands-on ML projects using real-world datasets.
- Practiced feature engineering, model evaluation, and EDA using Python.

EDUCATION & CERTIFICATIONS

- ✓ **MSc Data Science** — University of East London (2025–2027)
- ✓ **MBA (Strategic Management)** — Liverpool John Moores University (2024–2026)
- ✓ **Full-Stack Engineering** — Moringa (2022 - 2023)
- ✓ **Data Science** — Moringa (2020- 2021)
- ✓ **Monitoring and Evaluation** - Kenya Institute of Management (2019 - 2020)
- ✓ **BSC Health Sciences** - Mount Kenya University (2015)
- ✓ **AWS Certified Cloud Practitioner** (2025)